Pimpri Chinchwad Education Trust's

Pimpri Chinchwad College of Engineering and Research, Ravet

(An ISO 21001:2018 Certified Institute)

Department of Electronics & Telecommunication Engineering

ESDL Project Report

Course Code: VSE-281-ETC

ACADEMIC YEAR: 2025- 2026 **TERM : II**

Class: SE **Group No.** 13

Project Title: Library Seat Availability Monitoring System

Area of Project: Internet of Things (IoT)

Name of Faculty/Mentor: Mrs. Vijayalaxmi Kumbhar

Pimpri Chinchwad Education Trust's

Pimpri Chinchwad College of Engineering and Research, Ravet

Department of Electronics & Telecommunication Engineering

Certificate

This is to certify that, following students,

1. Isha Mishra
2. Parth Mishra
3. Mayuresh Mudase

has completed all the Term Work & Practical Work in the subject **Electronics Skill Development Lab (ESDL)** satisfactorily in the **Department of Electronics & Telecommunication Engineering** as prescribed by Savitribai Phule Pune University, in the academic year **2025 -2026.**

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Vision and Mission statements of the Institute

VISION

To be a globally recognized Institute of technological education and research for the holistic development of aspirants, through excellence in education, innovation and collaborations to fulfil the expectations of all stakeholders.

MISSIONS

1. To design and deliver state-of-art knowhow through experiential learning based on changing needs of industry and society worldwide, to ensure the employability and employment of each aspirant.
2. To enhance the collaborative partnership between Industry and Institute at national and international levels for commercializing and transferring the latest technological know-how towards societal, ethical and economic development
3. To achieve and sustain institute position as one of the topmost recognized and ranked institutes in technical and technological education

Vision and Mission statements of the Department

Vision:

To be a globally recognized center of excellence in Electronics and Telecommunication Engineering through quality education, innovation, and research, fostering professionals who contribute ethically and effectively to society and industry.

Mission:

1. To impart cutting-edge knowledge and practical skills in Electronics and Telecommunication Engineering through innovative and experiential learning to enhance employability and lifelong learning.
2. To foster research, innovation, and collaboration with industry and academia at national and international levels for technological advancement and societal benefit.
3. To cultivate ethical values, leadership qualities, and a sense of social and environmental responsibility among students for holistic professional development.

Program Educational Objective's

PEO1: To provide learning environment and hands on training in the domains of Electronics and Telecommunication engineering to achieve skill up-gradation of the human capital.

PEO2: To inculcate research and innovation capabilities in the Electronics and Telecommunication Engineering and allied interdisciplinary domains and adapt themselves to rapidly evolving technology.

PEO3: To impart moral, professional ethics and required skill set among the students so as to transform them into capable, adaptable and responsible citizens.

Program Specific Outcome's

A graduate of the E&TC Engineering Program will demonstrate -

PSO1: Graduate will be able to identify, design, prototype and test electronics and communication systems using software and hardware tools.

PSO2: Graduate will be able to develop and support systems based on embedded, automation, microwave, signal and image processing.

Knowledge and Attitude Profile (WK)

A Knowledge and Attitude Profile (KAP), often represented as WK (Knowledge and Attitude Profile) in some contexts, is a framework or assessment tool used to evaluate an individual's knowledge and attitudes related to a specific area, topic, or domain.

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Outcomes (POs) Engineering Graduates will be able to:

PO1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4	**Conduct Investigations of Complex Engineering Problems:** Using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO5	**Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6	**The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO7	**Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	**Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	**Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
PO10	**Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	**Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Course Objectives:

1. To impart knowledge about electronics system development
2. To make aware various tools and techniques for testing, simulation and PCB design
3. To acquaint industry standards for product development

Course Outcomes:

After successful completion of the course, learner will be able to:

VSE-281-ETC.1	Build application specific electronic circuit/system.
VSE-281-ETC.2	Use various measuring, debugging and EDA tools effectively.
VSE-281-ETC.3	Exercise prototype towards product development.

CO-PO-PSO Mapping-

CO\PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
VSE-281-ETC.1	2	2	3	2	2			2	2	2		2	
VSE-281-ETC.2	2	2	2	2	3			2	2		2	2	
VSE-281-ETC.3	2	2	3	2	2			3	2	3	2	2	
AVERAGE:	**2**	**2**	**2.66**	**2**	**2.33**			**2.33**	**2**	**2.5**	**2**	**2**	

Group Information:

Class: SE **Batch:** S2 **Group:** 13

Roll No.	PRN No.	Name of Student	Mobile No.
36	2402E01749	Isha Mishra	8446068527
37	2402E01753	Parth Mishra	7709871300
38	2402E01343	Mayuresh Mudase	9356662792

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Abstract

The Library Seat Management System is an IoT-based solution designed to efficiently monitor and manage seat occupancy in libraries. The system utilizes an ESP32 DevKit V1 microcontroller as the core processing unit, integrating both an IR sensor module and an FSR 402 (Force Sensitive Resistor) sensor to accurately detect whether a seat is occupied or vacant. The IR sensor is used to detect the presence of a user near the seat, while the FSR sensor measures applied pressure to confirm actual occupancy, reducing false detections.

The system provides real-time seat status through visual indicators using red and green LEDs, where red indicates an occupied seat and green represents availability. Supporting components such as resistors and capacitors ensure stable operation and signal conditioning. The ESP32's built-in Wi-Fi capability enables the system to transmit occupancy data to a cloud platform or web dashboard, allowing users to remotely check seat availability before entering the library.

This project aims to improve space utilization, reduce time spent searching for available seats, and enhance user convenience. Additionally, it can help library administrators monitor usage patterns and optimize seating arrangements. The proposed system is cost-effective, scalable, and easy to deploy, making it suitable for academic institutions and public libraries.

Overall, the Library Seat Management System demonstrates the practical application of embedded systems and IoT in solving real-world problems by providing a smart, automated, and efficient seating management solution.

Introduction

Modern academic and public libraries, managing seating availability efficiently has become a significant challenge due to increasing user demand and limited resources. Students and visitors often spend considerable time searching for vacant seats, leading to inconvenience, time wastage, and inefficient utilization of available space. Traditional methods of seat management, which rely on manual observation or fixed seating arrangements, fail to provide real-time and accurate information about seat occupancy.

To address these challenges, the proposed Library Seat Management System introduces a smart, automated solution based on embedded systems and Internet of Things (IoT) technology. The system is built around the ESP32 DevKit V1 microcontroller, which offers integrated Wi-Fi capabilities for seamless communication and data transmission. By incorporating an IR sensor module and an FSR 402 sensor, the system ensures reliable detection of seat occupancy through both proximity sensing and pressure measurement, thereby minimizing errors caused by false triggers.

The system also includes visual feedback using LED indicators to display seat status locally, enabling quick identification of available or occupied seats. Furthermore, the collected data can be transmitted wirelessly to a centralized platform, allowing users and administrators to monitor seat availability in real time.

This project emphasizes efficiency, scalability, and cost-effectiveness, making it suitable for deployment in libraries of varying sizes. By leveraging IoT and smart sensing technologies, the system aims to enhance user experience, optimize resource utilization, and contribute to the development of intelligent infrastructure in educational environments.

Initial Survey for Finalization of Title (Literature Survey)

Sr. No.	Author Name	Topic Name	Published In	Findings
1	Espressif Systems	ESP32 DevKit V1 Architecture and Wi-Fi Capabilities	Datasheet & Technical Manual	ESP32 provides dual-core processing, built-in Wi-Fi, and multiple GPIO pins suitable for real-time IoT systems.
2	Interlink Electronics	Force Sensitive Resistor (FSR 402)	Sensor Datasheet	FSR detects pressure variations effectively and is suitable for seat occupancy detection.
3	Arduino	Arduino IDE for Embedded Programming	Official Documentation	Arduino IDE enables easy programming and integration of sensors with ESP32.
4	Generic Manufacturer	IR Sensor Module Working Principle	Technical Datasheet	IR sensors detect objects using infrared reflection but may vary with environmental conditions.
5	Raj Kamal	Embedded Systems Design and Architecture	McGraw Hill	Provides fundamentals of embedded systems including hardware-software integration.
6	Arshdeep Bahga & Vijay Madisetti	Internet of Things: A Hands-On Approach	Academic Publication	Explains IoT architecture and real-world applications for smart systems.
7	Various Researchers	Smart Library System Using IoT	IEEE / Google Scholar	IoT-based systems improve library space utilization and reduce manual effort.
8	ThingSpeak / Blynk	Cloud Integration for IoT Systems	Online Platforms	Enable real-time monitoring, dashboards, and remote accessibility.

Problem Statement

In libraries, users often face difficulty in finding available seats due to the lack of a real-time seat monitoring system. This leads to time wastage, overcrowding in certain areas, and inefficient utilization of seating resources. Existing manual methods are unreliable and do not provide accurate occupancy information. Therefore, there is a need for a smart, automated system that can detect seat occupancy in real time and provide accurate availability status to improve user convenience and space management.

Objectives of Project

- To develop a real-time seat occupancy detection system:
  The primary objective is to design a system that can continuously monitor whether a library seat is occupied or vacant. This is achieved using sensors that provide instant feedback, ensuring up-to-date information at all times.
- To ensure accurate and reliable detection using multiple sensors:
  By integrating both an IR sensor (for proximity detection) and an FSR 402 sensor (for pressure sensing), the system aims to improve accuracy and reduce false readings, ensuring dependable performance in real-world conditions.
- To provide clear visual indication of seat status:
  The system uses red and green LEDs to display seat availability. This objective focuses on enabling users to quickly identify free or occupied seats without confusion or manual verification.
- To enable wireless communication and remote monitoring:
  Utilizing the ESP32 DevKit V1's built-in Wi-Fi, the system transmits real-time data to a cloud platform or web dashboard, allowing users and administrators to check seat availability remotely.
- To improve space utilization and reduce time wastage:
  The system is designed to help users quickly locate available seats, thereby reducing unnecessary movement and optimizing the use of library space.
- To design a cost-effective and scalable solution:
  Another objective is to create a system that is affordable and can be easily expanded to cover multiple seats or entire library sections without significant complexity.
- To implement a practical IoT-based application:
  The project aims to demonstrate the real-world application of embedded systems and IoT concepts, including sensor integration, data processing, and wireless communication..

Methodology

The development of the Library Seat Management System follows a structured approach consisting of hardware setup, sensor integration, data processing, and real-time communication.

1. Hardware Setup

Initially, the hardware components are assembled with the ESP32 DevKit V1 serving as the central controller. The IR sensor module and FSR 402 are interfaced with the ESP32 through appropriate GPIO pins.

The FSR sensor is connected using a voltage divider circuit with a 10 kΩ resistor to measure pressure variations accurately. The IR sensor is used for proximity detection. LEDs (red and green) are connected through 220 Ω resistors to indicate seat status. Additionally, capacitors of 0.1 µF and 10 µF are incorporated for noise filtering and power stabilization, ensuring reliable system operation.

2. Sensing Phase

In this phase, the system continuously reads data from both sensors. The IR sensor detects whether a person is present near the seat, while the FSR sensor confirms actual occupancy by sensing applied pressure.

A logical condition is implemented in the ESP32 program such that the seat is marked as "occupied" only when both sensors indicate presence, thereby minimizing false positives and improving accuracy.

3. Processing Phase

During processing, the ESP32 analyzes the inputs received from the sensors and determines the seat status. Based on this decision:

- The red LED is activated when the seat is occupied
- The green LED is activated when the seat is vacant

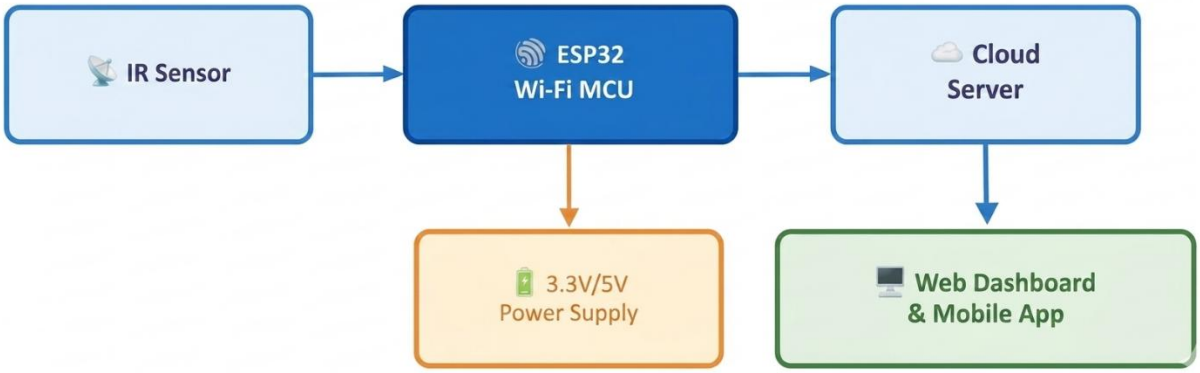
This provides immediate visual feedback to users.

4. Communication Phase

The ESP32 utilizes its built-in Wi-Fi module to transmit real-time occupancy data to a cloud server or web-based dashboard. This enables users and administrators to remotely monitor seat availability efficiently.

5. Testing and Calibration

Finally, testing and calibration are performed to optimize sensor thresholds and ensure accuracy. This step ensures reliable performance, minimizes errors, and enhances the overall efficiency of the system under real-world conditions.



Result as per objectives

The implementation of the Library Seat Management System achieved the defined objectives effectively, demonstrating accurate performance, reliability, and practical usability in a real-world scenario. The results obtained from testing and deployment are summarized below:

- Accurate Seat Occupancy Detection:
  The combined use of the IR sensor module and FSR 402 sensor ensured precise detection of seat occupancy. The dual-sensor approach minimized false positives, as the system only marked a seat as occupied when both proximity and pressure conditions were satisfied.
- Reduction in False Triggers:
  By integrating logic that validates input from both sensors, the system significantly reduced incorrect readings caused by temporary obstructions or environmental noise, improving overall reliability.
- Real-Time Status Indication:
  The LED indicators (red for occupied and green for vacant) responded instantly to changes in seat status, providing clear and immediate visual feedback to users within the library environment.
- Efficient Wireless Data Transmission:
  The ESP32 DevKit V1 successfully transmitted real-time seat availability data over Wi-Fi to a remote platform or dashboard. This enabled users to check seat status remotely and improved accessibility.
- Improved Space Utilization:
  The system reduced the time users spent searching for available seats, leading to better utilization of library space and minimizing overcrowding in specific areas.
- User Convenience and Experience:
  The automation of seat monitoring enhanced user experience by eliminating manual checking and providing a smart, user-friendly solution.
- System Stability and Performance:
  The inclusion of resistors and capacitors ensured stable circuit operation, and the system performed consistently during continuous testing without significant errors.

Conclusion and Future Scope

The Library Seat Management System successfully demonstrates the application of IoT and embedded systems in solving real-world problems related to seat availability and space utilization. By integrating the ESP32 DevKit V1 with IR and FSR sensors, the system provides accurate, real-time detection of seat occupancy while minimizing false readings. The use of LED indicators ensures immediate visual feedback, and wireless data transmission enhances accessibility by enabling remote monitoring. Overall, the system improves user convenience, reduces time wastage, and promotes efficient utilization of library resources in a cost-effective and scalable manner.

In terms of future scope, the system can be further enhanced by integrating a mobile application or web-based interface with advanced features such as seat reservation, user authentication, and notifications for seat availability. The addition of cloud data analytics can help administrators analyze usage trends and optimize seating arrangements. Moreover, expanding the system to cover multiple floors or entire institutions can improve scalability. Advanced sensors or AI-based occupancy detection using cameras could be incorporated for even higher accuracy. Energy-efficient designs and battery-powered modules can also be explored for flexible deployment. Thus, the project has strong potential for further development into a fully smart and intelligent library management solution.

References (Website/Books/Papers)

1. Espressif Systems, *ESP32 DevKit V1 Datasheet and Technical Reference Manual*. Available at: https://www.espressif.com
   → Used for understanding microcontroller architecture, Wi-Fi features, and GPIO interfacing.
2. Interlink Electronics, *FSR 402 Force Sensitive Resistor Datasheet*.
   → Used for studying pressure sensing characteristics and calibration techniques.
3. Arduino, *Arduino IDE Documentation*. Available at: https://www.arduino.cc
   → Used for ESP32 programming and sensor interfacing concepts.
4. *IR Sensor Module Datasheet (Generic IR Obstacle Sensor Documentation)*
   → Used for understanding the working principle and interfacing of IR-based detection.
5. Raj Kamal, *Embedded Systems: Architecture, Programming and Design*, McGraw Hill.
   → Used for fundamentals of embedded systems design.
6. Internet of Things: A Hands-On Approach
   → Used for IoT concepts, system design, and real-world applications.
7. *"Smart Library System Using IoT"*, IEEE / Google Scholar
   → Used for studying similar implementations and design approaches.
8. ThingSpeak / Blynk
   → Used for cloud integration and real-time data visualization.

Part List with Cost Estimation:

Sr. No	Component/Add on Educational courses	Quantity	Rate/ quantity	Total cost
1	ESP32 DevKit V1	1	440	440
2	1x15 Pin Headers (2.54 mm)	2	40	40
3	1x03 Pin Header	1	10	10
4	1x02 Pin Header	1	10	10
5	10 KΩ Resistor	1	5	5
6	220 Ω Resistors	2	10	10
7	LEDs (Red & Green)	2	4	4
8	0.1 µF Capacitor	1	5	5
9	10 µF Capacitor	1	5	5
10	IR Sensor Module	1	50	50
11	FSR 402 Sensor	1	220	220
Component/ Add on educational course cost total:				799